PNAS RESEARCH ARTICLE | CLASSIFICATION

Star Power and National TV Viewership in the NBA

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This manuscript was compiled on June 4, 2026

1. Introduction

This project builds directly on my prior work examining NBA style of play and audience engagement. That project found that star presence was a more reliable predictor of national TV viewership than stylistic factors such as pace, three-point rate, or fastbreak scoring. In other words, the question became less about whether fans dislike the modern style of NBA basketball and more about whether the right players are actually on the floor. This has become a major issue lately, as more and more star players are missing games due to injury, rest, and/or load management. This paper narrows in on that idea by moving from a general measure of star presence to player-specific and matchup-specific measures of star power to attempt to evaluate which stars have the largest impact on viewership.

The analysis uses three main approaches. First, I compare average viewership when each star played to average viewership when he did not play. Second, I estimate player-specific and matchup-specific OLS regression models, controlling for game-level factors such as team quality, betting spread, closing total, holiday status, and team valuation where appropriate. Finally, I use a random forest model to examine which players are most useful for predicting viewership. These models are not intended to prove that any one player causes a certain number of viewers to watch. Instead, they provide evidence about which stars and matchups are most strongly associated with larger national TV audiences.

Overall, this paper contributes to the broader discussion of NBA viewership by showing that star power is uneven. The results suggest that some players, especially Stephen Curry, Nikola Jokic, and LeBron James, are much more strongly associated with national TV audiences than others. This matters for how the NBA schedules games, markets its stars, and thinks about the regular-season product.

2. Related Work

Prior research on sports viewership suggests that audience demand is shaped by a combination of star power, team quality, competitiveness, and broadcast context. For this paper, the most relevant literature concerns the role of star players in viewership of NBA regular-season games.

The key idea for this project is the "superstar effect." Hausman and Leonard (1997) studied the economic value of superstars in the NBA and found that star players had meaningful effects on television ratings and league revenues (1). This result is central to the present paper because it suggests that viewers do not treat all NBA games equally. Instead, national audiences may be especially responsive to the presence of a small number of highly recognizable players. This paper builds on that framework by examining which specific stars are most associated with national TV viewership during the 2023–2024 regular season.

More recent work has connected this superstar effect to player absences. Reilly et al. (2023) examined missed games by NBA stars and found that star absences significantly reduce television audiences (2). This coincides with my previous work on this subject. This is especially relevant because modern NBA discourse often centers on load management and star availability. My paper extends this idea by not only asking whether stars matter in general, but also whether some stars matter more than others.

Another strand of literature focuses on uncertainty of outcome and game competitiveness. Solow et al. (2024) studied within-game uncertainty and found that competitive balance can affect television demand, particularly later in games (3). This matters because viewers may be less likely to stay engaged with games that become blowouts. For that reason, this paper includes betting spread as a control for expected competitiveness. The goal is to avoid confusing star power with the fact that certain stars may simply play in games that are expected to be more competitive.

Lastly, team quality and franchise popularity are also important. Foster et al. (2014) studied determinants of regional sports network ratings and found that factors such as recent success, ownership reputation, and All-Star presence helped predict ratings (4). This is relevant because some NBA teams have much larger fanbases and stronger national brands than others. A Lakers or Warriors game may draw viewers for reasons that go beyond any single player. To partially account for this, the present paper includes controls such as average team net rating and, in the matchup regressions, average team valuation.

3. Data

This paper investigates the possible relationship between NBA star playe
More specifically, it examines whether nationally televised regular-season g
associated with higher viewership, and whether specific star-versus-star matchups appear to draw more

viewers than other games. The dataset used in this paper comprises 195 regular-season NBA games from the 2023–2024 season. It is important to note that this dataset only includes games that have corresponding national TV viewership data. For this dataset, one observation equates to one game.

The data were collected from several different sources. National TV viewership data were collected from SportsMediaWatch. DartmouthGPT labeling was applied to categorize posts made on the website containing NBA viewership data, and then any pertinent information was pulled from those posts. Player participation data were collected manually for 2023–2024 NBA All-Stars to indicate whether each star played in each nationally televised game. Sportsbook data, including closing spreads and totals, were collected using an NBA betting dataset from Kaggle. Additional team-level controls, including team valuation and net rating, were added to account for differences in team popularity and team quality. The dependent variable measures traditional national TV viewership, measured as viewers in thousands (vwrs (000s)). The main regressors in this analysis are indicator variables for whether each 2023–2024 NBA All-Star played in a given game. These variables are coded as 1 if the player played and 0 if he did not. This allows the analysis to examine whether games involving certain stars are associated with higher national TV viewership. Because many individual all-star indicators appear in only a small number of games, some models remove players with five or fewer appearances in the dataset. This helps avoid estimating player effects from only a handful of observations.

The paper also considers star matchups by creating indicators for games in which two opposing stars both played. For example, a matchup indicator would equal 1 if Nikola Jokic and Stephen Curry both played in the same game and 0 otherwise. These indicators are used to explore whether certain star-versus-star matchups appear to draw higher viewership than other nationally televised games.

The subsequent analyses include numerous controls intended to account for factors influencing national TV viewership besides individual star popularity. Data were collected on the total number of all-stars playing in each game and the average net rating between the two teams playing in each game to account for the quality of the teams involved. Closing spread and closing total are included to account for expected competitiveness and expected scoring environment before the game. A control for the average valuation of the two teams playing in each game was also added. This variable is intended to capture some of the inherent popularity associated with certain franchises, as well as market size, which may make viewers more likely to watch a national broadcast regardless of the specific players involved.

A final disclaimer about the dataset is necessary. Due to data collection limitations, this analysis is restricted to the 195 regular-season games for which national TV viewership data were available. As a result, the findings should be interpreted as evidence from nationally televised games rather than from the full 2023–2024 NBA regular season. In addition, some player-specific and matchup-specific analyses rely on relatively small numbers of games, so those results are best interpreted as suggestive rather than definitive.

4. Methods

Little work was needed to clean the dataset, as it was largely identical to the data used in prior work. Two new pieces of information were added, however. First, a control variable for the average valuation of the two teams playing was added using a XXX merge join. Second, player indicator variables (e.g., lebron_james_played), were added for the 25 all-star selections in the 23-24 season and coded manually for all 195 games in the dataset. It should be noted that only players with 10 or more appearances in nationally televised games were included in the subsequent regressions and the random forest plot (n=19). The main dependent variable is national TV viewership, measured as the log of viewers in thousands (log(vwrs)).

4.1. Models of Analysis.

4.1.1. Descriptive Comparisons. The first set of analyses uses descriptive comparisons to examine whether games involving certain stars tend to draw higher viewership. I began with these descriptive analyses because they provide an intuitive overview of the relationship between star participation and audience size before introducing more complex statistical models. For each player, I compared the average viewership in games the player played to that in games the player did not play. This broad comparison helps identify which stars are most strongly associated with higher viewership across the league. The second descriptive analysis compares viewership within the games of a specific team when a certain player does or does not play. I include this team-level comparison because it helps account for differences in team popularity and market size, allowing for a clearer look at whether a player's presence is associated with changes in viewership among that team's games. These descriptive analyses do not isolate causal effects, but they provide a straightforward way to see whether certain stars are disproportionately associated with larger audiences and help motivate the subsequent regression analysis.

4.1.2. Player-Specific OLS Regressions. The second set of analyses uses player-specific ordinary least squares regression models. Instead of placing every player indicator into one regression at the same time, I ran separate regressions for each individual player. The general model is as follows:

$$log(vwrs) = \beta_0 + \beta_1 player\_played_i + \beta_2 avg\_net\_rating_i + \beta_3 spread_i + \beta_4 total_i + \beta_5 holiday_i + \epsilon_i \quad [1]$$

It should be noted that I do not include average team valuation in the main individual player regressions because team valuation is closely related to the popularity of the players themselves and may absorb part of the broader star-power effect I am trying to measure. In these models, the goal is to estimate whether games involving a given player are associated with higher viewership overall, controlling for game quality and context. However, I include average team valuation in the star matchup regressions because those models compare a small number of specific two-team matchups to all other games. In that setting, controlling for team valuation helps distinguish whether a matchup's viewership effect reflects the star-versus-star pairing itself rather than the general popularity of the franchises involved.

4.1.3. Star Matchup OLS Regressions. These regressions examine whether specific opposing star-matchups are associated with higher viewership. Matchup indicators were created in which both stars played against each other. For this analysis, only matchups between stars that had three or more appearances against each other were included (n=11). The following is the regression formula:

$$log(vwrs) = \beta_0 + \beta_1 Matchup_i + \beta_2 Controls_i + \epsilon_i \qquad [2]$$

Here, Matchup_i indicates whether a specific star matchup occurred in a given game. The controls included were avg_team_net_rating, avg_team_valuation_bil, holiday, total, and spread. These models are intended to test whether certain star-versus-star games appear to draw higher national TV audiences than other games, controlling for other game-level and team-level characteristics. Because these matchup indicators are based on a small number of games, the results of these specific models should be viewed with caution.

4.1.4. Random Forest Model. To supplement the regression analysis, I estimate a random forest model to identify which players are most useful for predicting national TV viewership. The outcome variable is viewership (log(vwrs)), and the predictors include All-Star player indicators along with selected game-level controls such as average net rating, spread, holiday status, and closing total. Players with fewer than ten appearances are excluded. I split the data into training and test sets, fit the random forest on the training data, and evaluate its performance on the held-out test set using (R^2), mean absolute error (MAE), and root mean squared error (RMSE). Next, prediction lift is calculated for each player, which compares the model's average predicted viewership when that player played to its average predicted viewership when he did not. These results are interpreted as predictive rather than causal, since high prediction lift may reflect a player's presence as well as broader factors that the model does not control for.

5. Results

5.1. Descriptive Comparisons. The first comparison examined the average viewers in games when each all-star played and compared that to the average viewers in games when each all-star did not play. Only players with at least 10 games missed and at least 10 games played were included. In the full-sample comparison, Stephen Curry had the largest positive difference, with games he played averaging about 599,000 more viewers than games he did not play. Jaylen Brown, Nikola Jokic, Jayson Tatum, LeBron James, and Anthony Davis also had relatively large positive differences. However, this comparison should be interpreted carefully because the "missed" category includes all other games in the dataset, not just games involving that player's team.

The second comparison attempted to softly control for team-specific effects by comparing viewership when each all-star played to viewership of games for their team when each all-star did not play. Only players with at least one missed game were included. See Table 1 for both sets of results. This team-specific comparison shows a similar pattern for some of the biggest stars. Nikola Jokic and Stephen Curry had the largest positive team-specific differences, with their teams' games drawing much higher average viewership when they played than when they missed. Kevin Durant and Damian Lillard also had large positive differences. At the same time, some players had negative differences, meaning their teams' games actually averaged more viewers when they did not play. Interestingly, Shai Gilgeous-Alexander has a strong negative association, which may at least in part be explained by his style of play. However, much of these effects likely reflect the small number of missed games for some players and the fact that national TV viewership is also shaped by opponent quality, matchup importance, day of the week, and other factors beyond individual player availability. Overall, the descriptive tables suggest that certain stars, especially Curry and Jokic, are associated with higher viewership, but these results should be interpreted as descriptive patterns rather than causal estimates.

5.2. Regression Results.

5.2.1. Player-Specific Regression Results. The player-specific regression results from Equation (1) show that Stephen Curry had the strongest positive association with viewership. Curry's coefficient was 0.585, which corresponds to an estimated 79.5% increase in viewership, controlling for average net rating, spread, total, and holiday status. This result was also statistically significant at the $p < 0.05$ level. LeBron James, Anthony Davis, and Nikola Jokic also had positive coefficients and relatively large estimated effects. LeBron James was associated with a 36.1% increase in viewership, Anthony Davis with a 31.4% increase, and Nikola Jokic with a 31.0% increase. Anthony Davis and Nikola Jokic were significant at the $p < 0.1$ level, while LeBron James was very close to that cutoff.

Table 1. Player Availability and Viewership Differences (000s)

Panel A: Full-Sample Differences

Player	Played	Missed	Avg. P	Avg. M	Diff.
S. Curry	25	170	1516.7	918.2	598.5
J. Brown	27	168	1273.2	950.2	323.0
N. Jokic	22	173	1277.1	959.1	318.0
J. Tatum	29	166	1241.5	951.9	289.6
L. James	23	172	1245.7	961.4	284.3
A. Davis	28	167	1183.7	963.3	220.4
L. Doncic	15	180	1145.7	982.4	163.3
D. Lillard	21	174	1138.8	977.6	161.2
G. Antetokounmpo	20	175	1095.2	983.5	111.7
J. Randle	10	185	1068.4	991.0	77.4
J. Brunson	14	181	1053.2	990.4	62.8
T. Maxey	19	176	1037.7	990.3	47.4
K. Durant	25	170	1023.4	990.8	32.7
B. Adebayo	15	180	992.5	995.2	-2.7
D. Booker	21	174	881.7	1008.6	-126.9
K. Leonard	19	176	869.6	1008.5	-138.9
P. George	20	175	837.3	1013.0	-175.7
A. Edwards	12	183	756.6	1010.6	-254.0
S. Gilgeous-Alexander	11	184	752.2	1009.5	-257.3

Panel B: Team-Specific Differences

Player	Team	Team G	Played	Missed	Avg. P	Avg. M	Diff.
N. Jokic	DEN	23	22	1	1277.1	247.0	1030.1
S. Curry	GSW	26	25	1	1516.7	495.0	1021.7
K. Durant	PHX	26	25	1	1023.4	392.0	631.4
D. Lillard	MIL	24	21	3	1138.8	587.0	551.8
J. Brown	BOS	29	26	3	1284.2	871.7	412.5
T. Maxey	PHI	24	19	5	1037.7	689.4	348.3
J. Embiid	PHI	24	9	15	1067.0	904.1	162.9
G. Antetokounmpo	MIL	24	20	4	1095.2	942.8	152.5
L. Doncic	DAL	16	15	1	1145.7	1007.0	138.7
L. James	LAL	31	23	8	1245.7	1150.8	94.9
T. Young	ATL	8	4	4	344.8	269.5	75.2
K. Towns	MIN	12	5	7	780.6	739.4	41.2
J. Randle	NYK	15	10	5	1068.4	1069.4	-1.0
K. Leonard	LAC	22	19	3	869.6	1022.0	-152.4
J. Brunson	NYK	15	14	1	1053.2	1286.0	-232.8
S. Gilgeous-Alexander	OKC	13	11	2	752.2	1072.5	-320.3
A. Davis	LAL	31	28	3	1183.7	1571.3	-387.7
D. Mitchell	CLE	11	5	6	604.4	1119.5	-515.1
P. George	LAC	22	20	2	837.3	1421.0	-583.7
D. Booker	PHX	26	21	5	881.7	1492.4	-610.7

Note: Panel A includes players with at least 10 games played and at least 10 games missed in the full sample. Panel B includes players with at least one missed game for their own team. Avg. P is average viewership when the player played, Avg. M is average viewership when the player missed, and Diff. is Avg. P minus Avg. M.

These results suggest that a small group of the league's most recognizable stars are most strongly associated with national TV viewership. This is consistent with the descriptive results, where Curry and Jokic also appeared near the top of the viewership difference tables. However, most player coefficients were not statistically significant. For example, players such as Jayson Tatum, Jaylen Brown, Luka Doncic, Damian Lillard, and Kevin Durant had coefficients that were not statistically distinguishable from zero in these models. Some players, including Devin Booker, Paul George, Anthony Edwards, and Shai Gilgeous-Alexander, had negative coefficients, though these estimates should not necessarily be interpreted as evidence that they reduce viewership. Instead, these negative estimates may reflect the specific games in which those players appeared, the teams they played against, or other factors not fully captured by the model.

Overall, the player-specific regressions provide partial support for the idea that individual stars are associated with higher national TV audiences. The strongest evidence is for Stephen Curry, followed by a smaller group of stars such as LeBron James, Anthony Davis, and Nikola Jokic. At the same time, the relatively modest R-squared values show that individual player availability explains only part of the variation in viewership. This makes sense because national TV viewership is likely shaped by many other factors, including team popularity, opponent quality, time slot, network, playoff implications, and broader fan interest.

5.2.2. Star Matchup Regression Results. The matchup regression results provide additional evidence that certain star-versus-star games are associated with higher viewership. These models controlled for average team net rating, average team valuation, holiday status, closing total, and closing spread. Including these controls helps account for differences in team quality, team popularity, game timing, expected scoring environment, and expected competitiveness. Even after including these controls, several matchups still had large positive associations with viewership.

The largest positive coefficient was for Nikola Jokic versus Paul George, which was associated with a 68.0% increase in viewership. Kevin Durant versus Luka Doncic and Luka Doncic versus Devin Booker were each associated with a 60.7% increase in viewership. Nikola Jokic versus Stephen Curry was also strongly associated with higher viewership, with an estimated 57.4% increase. These results suggest that some specific star matchups may draw larger audiences than individual star presence alone.

Several matchup results were statistically significant or close to significant. The Jokic-George, Durant-Doncic, Doncic-Booker, and Jokic-Curry matchups were statistically significant, while the Lillard-Brown and Tatum-Lillard matchups were close to the $p < 0.1$ level. This pattern suggests that nationally televised games may attract more viewers when they feature recognizable stars on both teams, especially when the matchup includes players with strong national followings.

However, these matchup results should be interpreted carefully. Each matchup indicator is based on a small number of games, usually only three or four. Because of this, a single high-viewership game can have a large effect on the estimated coefficient. The matchup regressions are therefore best understood as suggestive evidence rather than definitive proof that these pairings directly caused higher viewership. Still, the results fit the broader pattern of the paper: star power appears to matter for NBA national TV viewership, but its effect is uneven across players and matchups.

Table 2. Player and Matchup Regression Results

Panel A: Player Regressions

Player	Games	Coef.	Effect	p	R^2	N
S. Curry	25	0.585	79.5%	0.000	0.157	195
L. James	23	0.308	36.1%	0.100	0.109	195
A. Davis	28	0.273	31.4%	0.087	0.108	195
N. Jokic	22	0.270	31.0%	0.082	0.106	195
J. Brown	27	0.229	25.7%	0.207	0.100	195
J. Tatum	29	0.146	15.7%	0.445	0.096	195
J. Brunson	14	0.134	14.4%	0.664	0.096	195
B. Adebayo	15	0.122	13.0%	0.520	0.095	195
T. Maxey	19	0.101	10.6%	0.539	0.095	195
L. Doncic	15	0.095	10.0%	0.623	0.095	195
D. Lillard	21	0.054	5.6%	0.799	0.094	195
J. Randle	10	0.044	4.4%	0.912	0.094	195
G. Antetokounmpo	20	-0.024	-2.4%	0.915	0.094	195
K. Durant	25	-0.088	-8.4%	0.598	0.095	195
K. Leonard	19	-0.097	-9.2%	0.574	0.095	195
A. Edwards	12	-0.111	-10.5%	0.640	0.095	195
P. George	20	-0.119	-11.2%	0.473	0.096	195
D. Booker	21	-0.235	-20.9%	0.178	0.103	195
S. Gilgeous-Alexander	11	-0.327	-27.9%	0.146	0.103	195

Panel B: Matchup Regressions

Matchup	Games	Coef.	Effect	p	R^2	N
N. Jokic vs. P. George	3	0.519	68.0%	0.000	0.216	195
K. Durant vs. L. Doncic	3	0.474	60.7%	0.010	0.215	195
L. Doncic vs. D. Booker	3	0.474	60.7%	0.010	0.215	195
N. Jokic vs. S. Curry	4	0.453	57.4%	0.000	0.216	195
D. Lillard vs. J. Brown	4	0.403	49.6%	0.056	0.214	195
J. Tatum vs. D. Lillard	4	0.403	49.6%	0.056	0.214	195
G. Antetokounmpo vs. J. Brown	3	0.346	41.4%	0.227	0.210	195
G. Antetokounmpo vs. J. Tatum	3	0.346	41.4%	0.227	0.210	195
K. Durant vs. N. Jokic	3	0.310	36.3%	0.448	0.209	195
D. Lillard vs. J. Brunson	3	-0.850	-57.3%	0.428	0.208	195
G. Antetokounmpo vs. J. Brunson	3	-0.850	-57.3%	0.428	0.208	195
J. Brown vs. J. Randle	3	-0.972	-62.2%	0.147	0.211	195

Note: The dependent variable is log viewership. Effect converts the log coefficient into a percent change in viewership. Panel A includes player-specific regressions for players with at least 10 appearances. Panel B includes matchup-specific regressions for opposing star matchups with at least 3 games.

5.3. Random Forest Prediction Lift Results. The random forest model was used as a supplemental predictive model to examine which players were most useful for predicting national TV viewership. Unlike the regression models, the random forest does not estimate a direct coefficient for each player. Instead, it captures nonlinear relationships between player availability, selected game-level controls such as average net rating, spread, holiday status, and closing total. Prediction lift was then calculated by comparing the model's average predicted viewership when each player played to the model's average predicted viewership when that player did not play.

The model performed a bit better than a simple baseline model. The cross-validated R-squared was 0.144, meaning the model explained about 14.4 percent of the variation in viewership across games. The model had a cross-validated MAE of about 471,000 viewers and an RMSE of about 559,000 viewers. This improved on the baseline MAE by about 40,000 viewers and the baseline RMSE by about 45,000 viewers. These results suggest that the player indicators and game-level controls contain some predictive information, but also that a large amount of variation in national TV viewership remains unexplained by the model.

The prediction lift results again point to Stephen Curry as the strongest viewership predictor. Games in which Curry played had an average predicted viewership about 500,000 viewers higher than games in which he did not play. Jaylen Brown and Jayson Tatum also had large positive prediction lifts, at roughly 237,000 and 214,000 viewers, respectively. Nikola Jokic had the next highest lift, at about 186,000 viewers, followed by LeBron James at about 126,000 viewers. This is broadly consistent with the descriptive and regression results, which also showed Curry and Jokic near the top of the viewership results. See Figure 1 for a visualization of the results and the appendix for the complete set and model evaluation statistics.

However, the random forest results also differ from the regression results in some ways. For example, Jaylen Brown and Jayson Tatum have high prediction lift values even though their player-specific regression coefficients were not statistically significant. This likely reflects the fact that the random forest is capturing broader predictive patterns associated with Celtics games, rather than isolating a clean individual player effect. Similarly, some players had negative prediction lift values, including Shai Gilgeous-Alexander, Anthony Edwards, Kawhi Leonard, Tyrese Maxey, and Paul George. These negative values should not be interpreted as meaning these players directly reduce viewership. They may indicate that, in this dataset, the model predicted lower viewership for the games in which those players appeared after accounting for the other patterns it learned.

Overall, the random forest prediction lift results support the broader conclusion that some stars are more strongly associated with national TV viewership than others. Stephen Curry stands out most clearly across the descriptive comparisons, regression models, and prediction lift model. At the same time, the modest model performance and differences between the regression and random forest results show that national TV viewership is difficult to predict using player availability and basic game-level controls alone. Other factors, such as network, time slot, opponent, team popularity, and broader audience interest, likely play an important role.

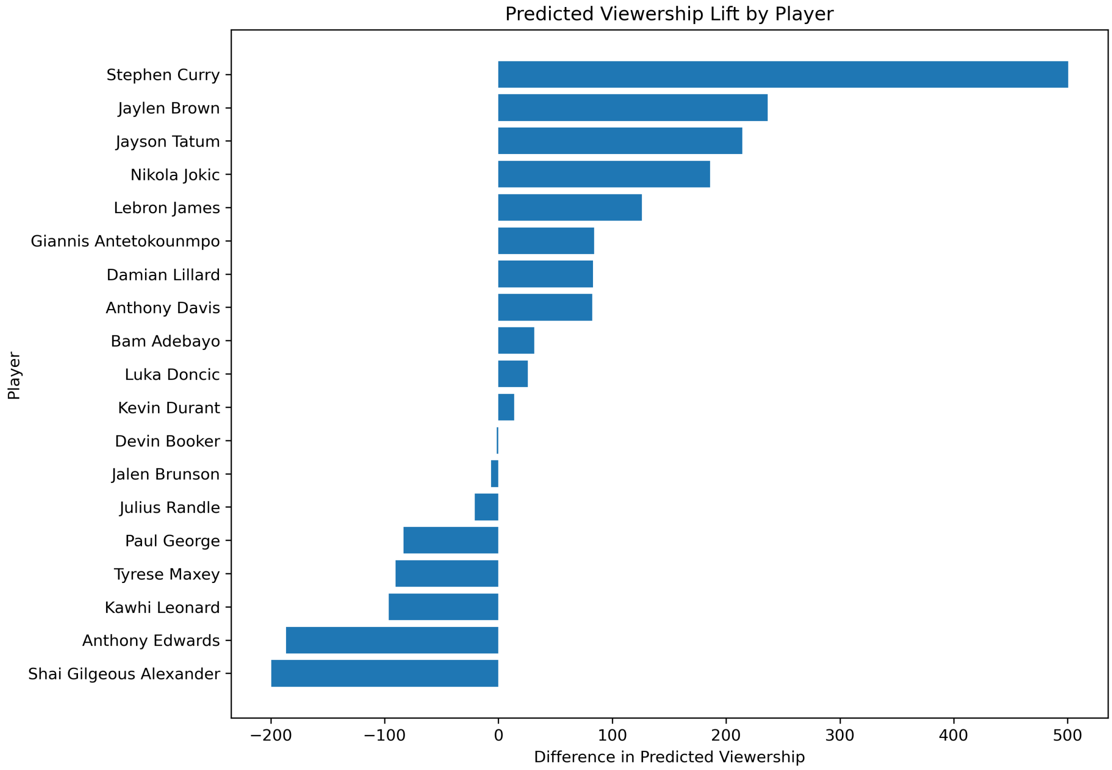

Fig. 1. Random Forest Prediction Lift by Player

6. Discussion

This paper finds that star power is associated with NBA national TV viewership, which coincides with my previous work on this subject. However, the results indicate this relationship is uneven across players and matchups. Across the descriptive comparisons, player-specific regressions, and random forest prediction lift results, Stephen Curry stands out as the clearest viewership driver. Nikola Jokic and LeBron James also appear consistently near the top of the results, while several other All-Stars show weaker or more inconsistent associations. The matchup regressions provide additional suggestive evidence that games featuring recognizable stars on both teams may draw larger audiences than games defined by individual star presence alone. Overall, the results support the idea that fans respond not just to the presence of stars generally, but to specific stars and specific star-versus-star matchups.

At the same time, these findings should be interpreted with caution. The analysis is limited to nationally televised regular-season games for which viewership data were available, so the results do not represent every NBA game from the 2023–2024 season. Because national TV games are already selected in advance, the sample likely overrepresents popular teams, major markets, and games expected to attract viewers. This makes it difficult to separate the effect of an individual player from the broader popularity of his team, opponent, network slot, or game context. The player-specific regressions also do not provide causal estimates. A positive coefficient for a player does not necessarily mean that he caused more people to watch; it means that games in which he played were associated with higher viewership after accounting for the included controls.

Another limitation is sample size. Several players and matchups appear in only a small number of nationally televised games, which makes those estimates sensitive to individual high-viewership or low-viewership games. This is especially important for the matchup regressions, where some indicators are based on only three or four games. In addition, the random forest model improved only modestly over a simple baseline, showing that player availability and basic game-level controls explain only part of national TV viewership. Other factors, such as broadcast network, time slot, day of week, opponent fanbase, playoff implications, injuries to non-All-Stars, and broader media attention likely matter as well.

Future work could improve this investigation by using multiple seasons of data, which would increase the number of observations for each player and matchup. It would also be useful to add controls for network, start time, day of week, team fixed effects, and opponent fixed effects. A different, but particularly useful, design could examine late injury scratches or unexpected player absences, since those situations would make it easier to isolate the effect of star availability from the league's scheduling decisions. Even with these limitations, the main takeaway is clear: NBA viewership is likely not driven by star power equally. Certain players, especially Curry, LeBron, and Jokic, appear to have a much stronger relationship with national TV audiences, which has important implications for scheduling, marketing, and the value of regular-season star availability.

7. References

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8. Website Link

https://qss20-nba-engagement.vercel.app/

9. Github Repo Link

https://github.com/jqmunro41/qss20-nba-engagement

10. Appendix

10.1. Random Forest Full Results & Model Evaluation Statistics.

Table 3. Random Forest Prediction Lift by Player

Player	Appearances	Pred. When Played	Pred. When Not Played	Prediction Lift	Actual Difference
Stephen Curry	25	1440.8	940.4	500.4	598.5
Jaylen Brown	27	1208.5	971.8	236.7	323.0
Jayson Tatum	29	1186.9	972.7	214.2	289.6
Nikola Jokic	22	1169.6	983.6	186.0	318.0
LeBron James	23	1115.8	989.7	126.1	284.3
Giannis Antetokounmpo	20	1080.2	995.9	84.2	111.7
Damian Lillard	21	1078.6	995.7	82.9	161.2
Anthony Davis	28	1075.3	992.7	82.5	220.4
Bam Adebayo	15	1033.6	1002.2	31.4	-2.7
Luka Doncic	15	1028.4	1002.6	25.8	163.3
Kevin Durant	25	1016.7	1002.8	13.9	32.7
Devin Booker	21	1003.3	1004.7	-1.5	-126.9
Jalen Brunson	14	998.7	1005.0	-6.4	62.8
Julius Randle	10	984.9	1005.7	-20.7	77.4
Paul George	20	929.7	1013.1	-83.4	-175.7
Tyrese Maxey	19	923.0	1013.4	-90.4	47.4
Kawhi Leonard	19	917.6	1014.0	-96.4	-138.9
Anthony Edwards	12	829.4	1016.1	-186.7	-254.0
Shai Gilgeous-Alexander	11	816.2	1015.9	-199.7	-257.3

Note: Viewership is measured in thousands (000s). Prediction lift is the difference between the model's average predicted viewership when the player played and when the player did not play. Only players with at least 10 appearances are included.

Table 4. Random Forest Model Evaluation

Metric	Value
Cross-validated R^2	0.144
Cross-validated MAE	470.849
Cross-validated RMSE	558.791
Baseline MAE	510.664
Baseline RMSE	603.972
MAE Improvement Over Baseline	39.815
RMSE Improvement Over Baseline	45.181
Observations	195

Note: MAE and RMSE are measured in thousands of viewers. The baseline model predicts average viewership without using the player and game-level predictors.

10.2. Agentic Analysis. I chose to use ChatGPT as my AI assistant for this project. It worked pretty well. For the code, I used it to help construct scripts for the regressions and prediction lift models. I also used it to generate my website and write the README for my Github Repo. The assistant made few errors, and I accepted most of what it generated because it was working successfully. There were some issues with the website creation, but I kept asking better questions which led to a successful product. Overall, it was a good experience using ChatGPT for help with this project.

10.2.1. Transcripts. Q: Write code for regression analysis on each player individually

A:

```
    import pandas as pd
import numpy as np
import statsmodels.formula.api as smf

df = pd.read_csv("ratings_final_with_team_valuations_net_rating_filled.csv")

# log outcome
df["log_vwrs"] = np.log(df["vwrs"])

star_cols = [col for col in df.columns if col.endswith("_played")]

appearance_counts = df[star_cols].sum()
star_cols_over_5 = appearance_counts[appearance_counts > 5].index.tolist()

results = []

for player in star_cols_over_5:

    formula = (
        f"log_vwrs ~ {player} + "
        "avg_team_valuation_bil + holiday + spread + total + avg_net_rating + C(network)"
    )

    model = smf.ols(formula=formula, data=df).fit(cov_type="HC3")

    coef = model.params[player]

    # Convert log coefficient into approximate percent effect
    percent_effect = (np.exp(coef) - 1) * 100

    results.append({
        "player": player.replace("_played", "").replace("_", " ").title(),
        "appearances": int(appearance_counts[player]),
        "coef_log": coef,
        "percent_effect": percent_effect,
        "p_value": model.pvalues[player],
        "r_squared": model.rsquared,
        "n_obs": int(model.nobs)
    })

results_df = pd.DataFrame(results)
results_df = results_df.sort_values("percent_effect", ascending=False)

print(results_df)
```

Q: can I justify not included avg_team_valuation_bil in the player specific regressions but including it in the matchup ones?

A: Yes, you can justify that, but you need to be clear that the two models are asking slightly different questions.

For the individual player regressions, it is defensible to leave out $avg_{t}eam_{v}aluation_{b}il$ because team valuation may absorb part of the star's overall appeal. For example, LeBron's popularity is tied to the Lakers brand, Steph's to the Warriors brand, and Tatum's to the Celtics brand. If you include valuation, you are asking a stricter question:

Does this player increase viewership beyond the general popularity of his team?

That may be too strict if your research question is about whether certain stars are associated with higher viewership overall.

For the matchup regressions, including $avg_{t}eam_{v}aluation_{b}il$ makes more sense because you are comparing specific two-team games to the rest of the dataset. In that case, you want to make sure a Jokic-Curry effect is not just a Warriors/Nuggets brand effect, or a KD-Luka effect is not just a Suns/Mavericks brand effect.

You could write it like this in your methods section:

I do not include average team valuation in the main individual player regressions because team valuation is closely related to the popularity of the players themselves and may absorb part of the broader star-power effect I am trying to measure. In those models, the goal is to estimate whether games involving a given player are associated with higher viewership overall, controlling for game quality and context. However, I include average team valuation in the star matchup regressions because those models compare a small number of specific two-team matchups to all other games. In that setting, controlling for team valuation helps distinguish whether a matchup's viewership effect reflects the star-versus-star pairing itself rather than the general popularity of the franchises involved.

That is a reasonable distinction.

Just make sure you are consistent in how you describe the results:

Without $avg_{t}eam_{v}aluation_{b}il$, the individual player regressions show overall player association with viewership.

With $avg_{t}eam_{v}aluation_{b}il$, the matchup regressions show matchup association with viewership net of team brand/popularity

Q: for the random forest plot, could I use something different than permutation importance as the x axis? I still want to show which players are best for predicitnv views?

A: Yes. You have a few reasonable options. For your project, I would probably use drop-column importance or prediction lift instead of raw permutation importance, because they are easier to explain.

Best option: drop-column importance

This asks:

How much worse does the model perform when this player is removed?

The x-axis can be:

Increase in Test MAE When Player Is Removed

That is very intuitive. Larger values mean the player helps the model predict viewership more.

Another good option: prediction lift

This asks:

According to the random forest, how much higher are predicted views when this player plays?

This is not really "importance," but it is very easy to understand.

This x-axis is:

Difference in Predicted Viewership

That may be the easiest graph for readers.

Which one should you use?

I would use drop-column importance if your goal is:

Which players are most useful for predicting viewership?

I would use prediction lift if your goal is:

Which players are associated with higher predicted viewership?

For your paper, drop-column importance is more defensible as a random forest feature-importance plot. Prediction lift is more intuitive, but it is closer to a descriptive predicted comparison.

Q: Write a script for the prediction lift

A:

```
    import pandas as pd
import matplotlib.pyplot as plt

from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, r2_score, mean_squared_error

# Load data
df = pd.read_csv("Final Datasets/qss20_finalds.csv")

# Player indicator columns
star_cols = [col for col in df.columns if col.endswith("_played")]

# Optional: only keep players with at least 10 appearances
appearance_counts = df[star_cols].sum()
star_cols = appearance_counts[appearance_counts >= 10].index.tolist()

# Controls
control_cols = [
    "avg_net_rating",
    "spread",
    "holiday",
    "total"
]
```

```
# Keep only controls that exist
control_cols = [col for col in control_cols if col in df.columns]

# Model dataframe
model_df = df[["vwrs"] + star_cols + control_cols].dropna()

# X and y
X = model_df[star_cols + control_cols]
y = model_df["vwrs"]

# Train/test split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
    random_state=1
)

# Fit random forest
rf = RandomForestRegressor(
    n_estimators=500,
    random_state=1
)

rf.fit(X_train, y_train)

# Predict on test set
y_pred = rf.predict(X_test)

# Model evaluation
test_r2 = r2_score(y_test, y_pred)
test_mae = mean_absolute_error(y_test, y_pred)
test_rmse = mean_squared_error(y_test, y_pred) ** 0.5

# Baseline model: always predict the training-set average
baseline_pred = [y_train.mean()] * len(y_test)

baseline_mae = mean_absolute_error(y_test, baseline_pred)
baseline_rmse = mean_squared_error(y_test, baseline_pred) ** 0.5

mae_improvement = baseline_mae - test_mae
rmse_improvement = baseline_rmse - test_rmse

print("Model evaluation:")
print("Test R-squared:", test_r2)
print("Test MAE:", test_mae)
print("Test RMSE:", test_rmse)

print("\nBaseline comparison:")
print("Baseline MAE:", baseline_mae)
print("Baseline RMSE:", baseline_rmse)
print("MAE improvement over baseline:", mae_improvement)
print("RMSE improvement over baseline:", rmse_improvement)

# Create copy of test data with predictions
test_results = X_test.copy()
test_results["actual_vwrs"] = y_test
test_results["predicted_vwrs"] = y_pred

# Calculate prediction lift for each player
lift_results = []

for player in star_cols:
```

```
    played_pred = test_results.loc[test_results[player] == 1, "predicted_vwrs"]
    not_played_pred = test_results.loc[test_results[player] == 0, "predicted_vwrs"]

    played_actual = test_results.loc[test_results[player] == 1, "actual_vwrs"]
    not_played_actual = test_results.loc[test_results[player] == 0, "actual_vwrs"]

    # Only calculate if player appears at least once in test set
    if len(played_pred) > 0 and len(not_played_pred) > 0:

        prediction_lift = played_pred.mean() - not_played_pred.mean()
        actual_difference = played_actual.mean() - not_played_actual.mean()

        lift_results.append({
            "player": player.replace("_played", "").replace("_", " ").title(),
            "total_appearances": int(appearance_counts[player]),
            "test_appearances": int(test_results[player].sum()),
            "avg_predicted_when_played": played_pred.mean(),
            "avg_predicted_when_not_played": not_played_pred.mean(),
            "prediction_lift": prediction_lift,
            "avg_actual_when_played": played_actual.mean(),
            "avg_actual_when_not_played": not_played_actual.mean(),
            "actual_difference": actual_difference
        })

lift_df = pd.DataFrame(lift_results)

# Sort by highest prediction lift
lift_df = lift_df.sort_values("prediction_lift", ascending=False)

print("\nPrediction lift results:")
print(lift_df)

# Plot top 15 players
top_players = lift_df.head(15)

plt.figure(figsize=(10, 7))

plt.barh(
    top_players["player"],
    top_players["prediction_lift"]
)

plt.xlabel("Difference in Predicted Viewership")
plt.ylabel("Player")
plt.title("Predicted Viewership Lift by Player")
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```

Q: Turn regression results into nice table for website

A:

I turned it into a cleaner presentation-style regression table with formatted coefficients, percent effects, p-values, significance markers, R^2, and N.

Files:

Download PNG table Download HTML table Download cleaned CSV table

The main takeaway from the table is that Stephen Curry is the only clearly statistically significant player effect in these results: about +79.5% estimated viewership, with p ¡ 0.001. Anthony Davis and Nikola Jokic are positive and near the 10% significance threshold, but not conventionally significant at p ¡ 0.05.

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  font-weight: bold;
  margin-bottom: 10px;
}
```

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  text-align: center;
  border-bottom: 2px solid black;
  padding: 6px;
}
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  padding: 6px;
}
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  font-size: 13px;
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</style>
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  </thead>
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      <td id="T_fae1e_row0_col4" class="data row0 col4" ><0.001</td>
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      <td id="T_fae1e_row2_col5" class="data row2 col5" >†</td>
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```

```
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      <td id="T_fae1e_row4_col3" class="data row4 col3" >+25.7%</td>
      <td id="T_fae1e_row4_col4" class="data row4 col4" >0.207</td>
      <td id="T_fae1e_row4_col5" class="data row4 col5" ></td>
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    </tr>
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      <td id="T_fae1e_row5_col0" class="data row5 col0" >Jayson Tatum</td>
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      <td id="T_fae1e_row5_col2" class="data row5 col2" >0.146</td>
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      <td id="T_fae1e_row6_col0" class="data row6 col0" >Jalen Brunson</td>
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```

```
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  <td id="T_fae1e_row11_col4" class="data row11 col4" >0.912</td>
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  <td id="T_fae1e_row11_col6" class="data row11 col6" >0.094</td>
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  <td id="T_fae1e_row13_col1" class="data row13 col1" >25</td>
  <td id="T_fae1e_row13_col2" class="data row13 col2" >-0.088</td>
  <td id="T_fae1e_row13_col3" class="data row13 col3" >-8.4%</td>
  <td id="T_fae1e_row13_col4" class="data row13 col4" >0.598</td>
  <td id="T_fae1e_row13_col5" class="data row13 col5" ></td>
  <td id="T_fae1e_row13_col6" class="data row13 col6" >0.095</td>
  <td id="T_fae1e_row13_col7" class="data row13 col7" >195</td>
</tr>
<tr>
  <td id="T_fae1e_row14_col0" class="data row14 col0" >Kawhi Leonard</td>
  <td id="T_fae1e_row14_col1" class="data row14 col1" >19</td>
  <td id="T_fae1e_row14_col2" class="data row14 col2" >-0.097</td>
  <td id="T_fae1e_row14_col3" class="data row14 col3" >-9.2%</td>
  <td id="T_fae1e_row14_col4" class="data row14 col4" >0.574</td>
  <td id="T_fae1e_row14_col5" class="data row14 col5" ></td>
  <td id="T_fae1e_row14_col6" class="data row14 col6" >0.095</td>
  <td id="T_fae1e_row14_col7" class="data row14 col7" >195</td>
</tr>
<tr>
  <td id="T_fae1e_row15_col0" class="data row15 col0" >Anthony Edwards</td>
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  <td id="T_fae1e_row15_col2" class="data row15 col2" >-0.111</td>
  <td id="T_fae1e_row15_col3" class="data row15 col3" >-10.5%</td>
  <td id="T_fae1e_row15_col4" class="data row15 col4" >0.640</td>
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    </tr>
    <tr>
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      <td id="T_fae1e_row17_col7" class="data row17 col7" >195</td>
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Link to Website Creation transcript:
https://chatgpt.com/share/6a219b93-8ec4-83ea-baee-96408c71a69f
Link to README creation transcript:
https://chatgpt.com/share/6a219c04-cca8-83ea-8aa7-430b9928f20d